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BME 195 Gene Editing
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The study 'Restoration of Visual Function in Adult Mice with an Inherited Retinal Disease via Adenine Base Editing' by Suh et al. explores the use of adenine base editors (ABEs) to address genetic mutations associated with Leber Congenital Amaurosis, an inherited retinal disease that leads to severe visual impairment (Suh et al.).

To test their approach, the researchers employed a lentiviral vector to deliver ABE and guide RNA to the retinal pigment epithelium (RPE) cells in a mouse model (rd12) carrying the Rpe65 mutation. By precisely converting a single base in the genetic sequence, the ABE corrected up to 29% of the targeted alleles, restoring RPE65 protein expression and its associated enzymatic activity. This correction reactivated the visual cycle, which was evidenced by the production of the chromophore 11-cis-retinal, and significantly improved retinal function. These improvements were confirmed through various assays, including electroretinography (ERG) and optomotor response tests, which showed that treated mice exhibited near-normal visual performance. The researchers demonstrated the precision of the ABE technology, with minimal off-target effects and bystander editing.

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While their study focuses on inherited retinal diseases, the same principles of base editing can be applied to other genetic disorders with similarly well-defined mutations. One such disorder is Sickle Cell Disease (SCD), a severe and prevalent monogenic disorder caused by a single nucleotide mutation in the β-globin (HBB) gene. SCD results from a Glu6Val substitution (GAG→GTG). A method of ABE to correct SCD is already being researched. For instance, research by Newby et al. achieved significant therapeutic outcomes by delivering ABE components into hematopoietic stem and progenitor cells (HSPC) via electroporation, followed by transplantation into humanized mouse models (Newby et al.). While this ex vivo approach has shown promise, it can still be improved. High costs, complex cell handling, and the requirement for transplantation are still challenges this method faces. I propose modifying the delivery system to technologies like lipid nanoparticles (LNPs). This delivery system offers the potential for efficient systemic or in vivo editing, simplifying the therapeutic process while maintaining high efficacy.

LNPs have revolutionized nucleic acid delivery, particularly in the field of mRNA therapeutics, as evidenced by their critical role in the success of COVID-19 vaccines. Beyond their application in vaccines, LNPs represent a transformative advancement in gene editing delivery systems, offering significant advantages over traditional methods such as electroporation. Electroporation, while effective for ex vivo editing, requires intricate and resource-intensive cell-handling procedures, including the isolation of HSPCs, electroporation to introduce genetic editing components, and subsequent transplantation back into the patient. These steps impose logistical and technical barriers, limiting scalability and accessibility, especially in clinical settings (Raguram et al.) (Zheng et al.).

In contrast, LNPs enable direct, in vivo delivery of therapeutic payloads, bypassing the need for ex vivo cell manipulation entirely as visualized in Figure 1. This streamlined approach eliminates the complexities associated with electroporation, providing a patient-friendly alternative that reduces treatment burden. LNPs achieve this efficiency through their high encapsulation capacity, which protects nucleic acids such as mRNA during systemic circulation. Ionizable lipids, a critical component of LNPs, ensure stability in circulation by remaining neutral at physiological pH while adopting a positive charge in acidic endosomal environments. This pH-sensitive behavior facilitates endosomal escape, enabling an efficient cytoplasmic release of therapeutic payloads (Raguram et al.).

LNPs further address key limitations of electroporation by avoiding physical stress and cytotoxicity, which can compromise HSPC viability. Through careful engineering, LNP particle sizes can be tuned to an optimal range of 50–100 nm, improving tissue penetration and ensuring more precise delivery to target cells (Zheng et al.). Surface modifications, such as the conjugation of CD34-specific ligands, enhance targeting specificity to HSPCs, minimizing off-target effects and reducing systemic toxicity. By enabling systemic in vivo delivery, LNPs eliminate the need for cell isolation and transplantation, simplifying the treatment process for ABE-based therapies targeting the Glu6Val mutation in SCD (Hou et al.).

To validate advanced delivery methods for ABE in correcting the Glu6Val mutation associated with SCD, this study will employ a three-phase experiment approach: in vitro optimization, in vivo validation, and comparative evaluation. These will aim to systematically optimize LNPs as an innovative delivery system, building on their established success in therapeutic RNA and protein delivery.

In the first phase, in vitro optimization and validation will focus on improving the delivery efficiency of LNPs for ABE components into CD34+ HSPCs isolated from healthy donors or SCD patient samples. Optimizations will include modifying lipid compositions, such as adjusting the ratios of ionizable lipids, cholesterol, and PEG-lipids, to enhance encapsulation efficiency, intracellular delivery, and endosomal escape (Raguram et al.). Techniques such as tuning particle size to between 50–100 nm, which improves tissue penetration and cellular uptake, will also be employed (Zheng et al.). Furthermore, surface modifications, including conjugation with ligands that specifically target CD34+ HSPCs, will be explored to improve cell specificity and minimize off-target effects. Editing efficiency will be quantified using PCR and next-generation sequencing (NGS), while cell viability and proliferation will be assessed using flow cytometry and colony-forming unit assays. Functional validation will involve differentiating edited HSPCs into erythroid cells and analyzing β-globin expression using high-performance liquid chromatography (HPLC). Precision will be evaluated by analyzing off-target effects with genome-wide tools like GUIDE-seq (Zheng et al.).

The second phase involves in vivo validation using a humanized mouse model of SCD expressing the Glu6Val mutation. Optimized LNPs encapsulating ABE mRNA and sgRNA will be administered intravenously to evaluate systemic delivery efficiency (Zheng et al.) as demonstrated in Figure 2. Editing efficiency will be measured by quantifying corrected alleles in the bone marrow and peripheral blood using NGS. Therapeutic efficacy will be assessed by monitoring phenotypic recovery, such as reductions in sickled red blood cells, improved hemoglobin levels, and normalization of spleen size and vascular occlusion. Safety will be evaluated by measuring immune responses, such as cytokine profiles, and by assessing off-target effects through GUIDE-seq. These experiments aim to demonstrate the feasibility of achieving therapeutically relevant levels of genome correction with minimal immune activation or bystander effects.

The final phase will compare the performance of LNPs against electroporation, which currently serves as the benchmark for ex vivo delivery of ABE into HSPCs. Metrics for comparison will include editing efficiency, as determined by the percentage of corrected alleles, and safety, assessed through genome-wide profiling of off-target effects and immune responses. Unlike ex vivo approaches that require complex cell handling and re-transplantation, in vivo, delivery via LNPs offers the advantage of reducing patient burden and improving accessibility (Raguram et al.) (Zheng et al.). These comparisons will provide critical insights into the practicality of transitioning from traditional delivery methods to advanced, scalable platforms capable of addressing the challenges associated with gene editing therapies for SCD.

Figures

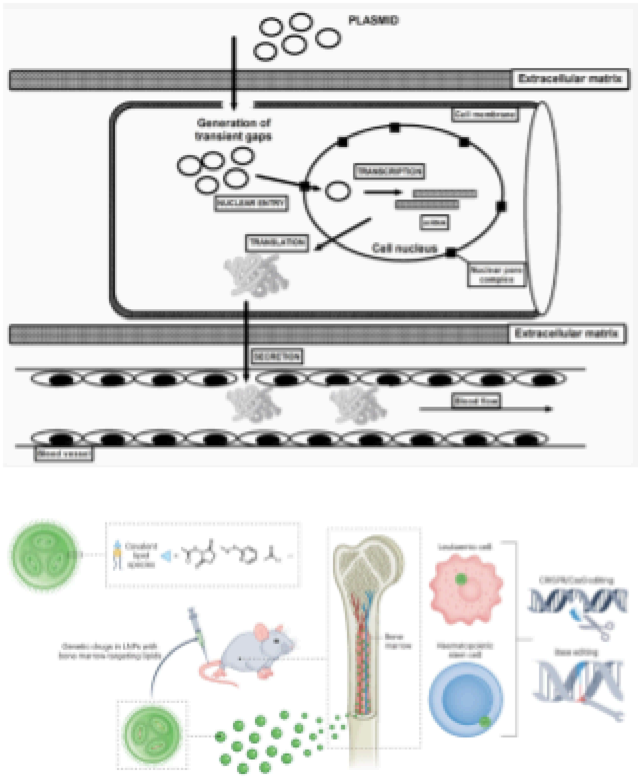

Figure 1:

Top Panel: *Schematic diagram of electroporation-based plasmid DNA delivery. Plasmid DNA (pDNA) must traverse biological barriers, including the extracellular matrix, cell membrane, cytoplasm, and nuclear membrane, to reach the nucleus. Application of an electric current creates transient gaps in the cell membrane, allowing pDNA entry into the cytoplasm. However, this process exposes pDNA to cytoplasmic nucleases, leading to degradation risks before nuclear entry. Once inside the nucleus, the pDNA is transcribed to produce the desired protein or silences target genes for therapeutic effects. Despite its effectiveness, electroporation introduces logistical complexity and risks of cell viability reduction. (Sokolowska et. al)*

Bottom Panel: *Mechanism of lipid nanoparticle (LNP)-based delivery for genetic editing. Functionalized LNPs incorporating reactive head groups and conventional lipids (phospholipids, ionizable lipids, cholesterol, PEG-lipids) encapsulate genetic drugs such as CRISPR–Cas9 or base editors. Following intravenous administration in mice, LNPs target hematopoietic stem cells (HSCs) in the bone marrow, allowing precise and efficient in vivo genetic editing. This method streamlines the editing process by eliminating ex vivo cell manipulation and minimizing risks to cell viability, positioning LNPs as a transformative alternative to electroporation. (Bitounis et. al)*

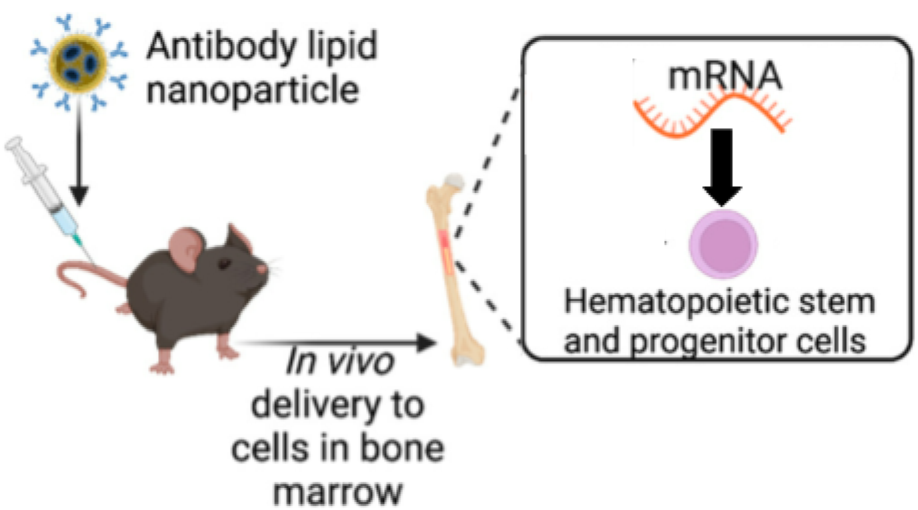

Figure 2: Schematic representation of the proposed delivery system for adenine base editors (ABEs) using lipid nanoparticles (LNPs). Antibody-functionalized LNPs are administered via intravenous injection, targeting hematopoietic stem and progenitor cells (HSPCs) in the bone marrow. The LNPs encapsulate mRNA payloads, enabling in vivo editing of genetic mutations directly within the target cells, as illustrated by the intracellular delivery of mRNA to HSPCs. (via Biorender)

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